

31 Halogen compounds

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Aryl halides are less reactive in nucleophilic substitution than haloalkanes because carbon-halogen bonding to the aromatic ring is strengthened by resonance and the sp^2 carbon environment. Haloalkane elimination competes with substitution; solvent, temperature and substrate structure influence outcome.

Worked example: follow the reasoning

Aqueous OH^- with a primary bromoalkane tends to form an alcohol; hot ethanolic OH^- favours alkene formation.

Quick check · multiple choice

A Which conditions promote haloalkane elimination?

[1]

A Cold aqueous OH^-

B Aqueous AgNO_3

C Hot ethanolic OH^-

D Dry HCl only

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Describe two changes to favour elimination rather than substitution of 2-bromobutane and give two possible alkene products.

[3]

Answers and explanation

Quick check: C. Hot ethanolic OH^-

Question 1 [3 marks]

Use ethanolic rather than aqueous base and heat [2]; but-1-ene and but-2-ene [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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